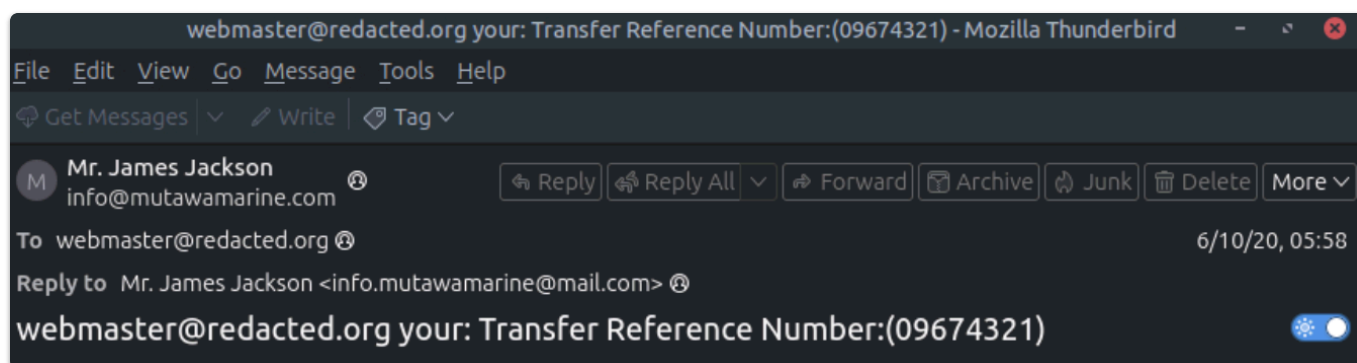


Scenario Overview :

A sales executive at Greenholt PLC has reported a suspicious email received from a known customer. The message raised several red flags: a generic greeting, an unexpected request for a money transfer, and an unsolicited attachment. According to the employee, this behavior does not align with the customer's usual communication style. Concerned that the email may be malicious

Email header

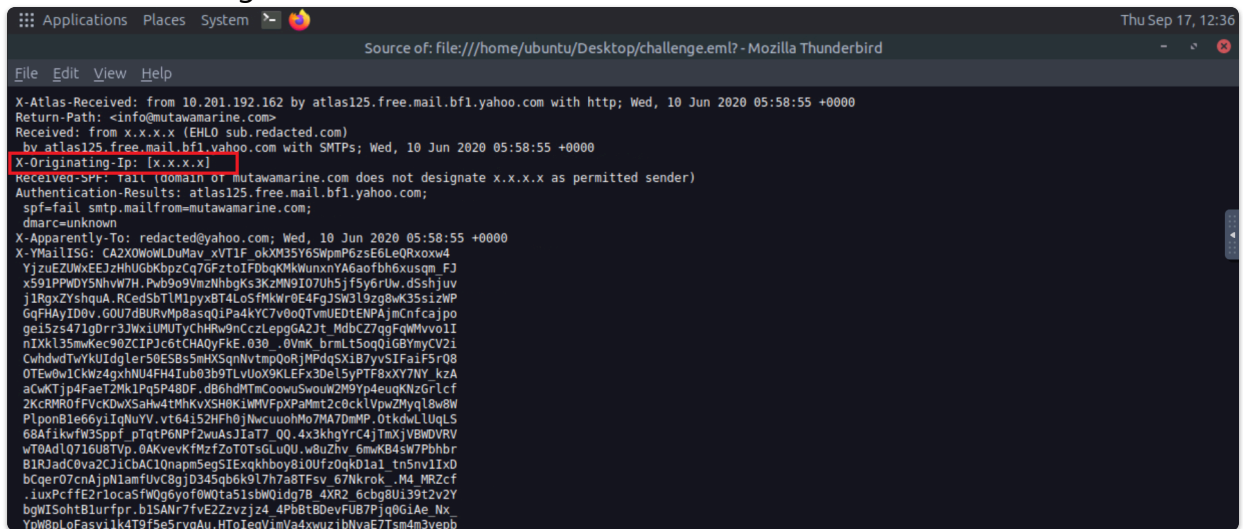


Email Content

Good day webmaster@redacted[.]org ,
As instructed, funds has been transferred to your account this morning via SWIFT.
Details are as below and a receipt of payment is attached.
Interbank Transfer Reference Number: 09674321
Transaction Status: Successful
Transaction Date / Time: 10-06-2020 09:18:55
Transaction Description: Balance / Final Payment
From Account: 3105234819
Amount: 149,650
Currency: usd
Bank Charges: \$ 146.05
Best regards,
Mr. James Jackson
Accounts Payable
SEC MARINE SERVICES PTE LTD

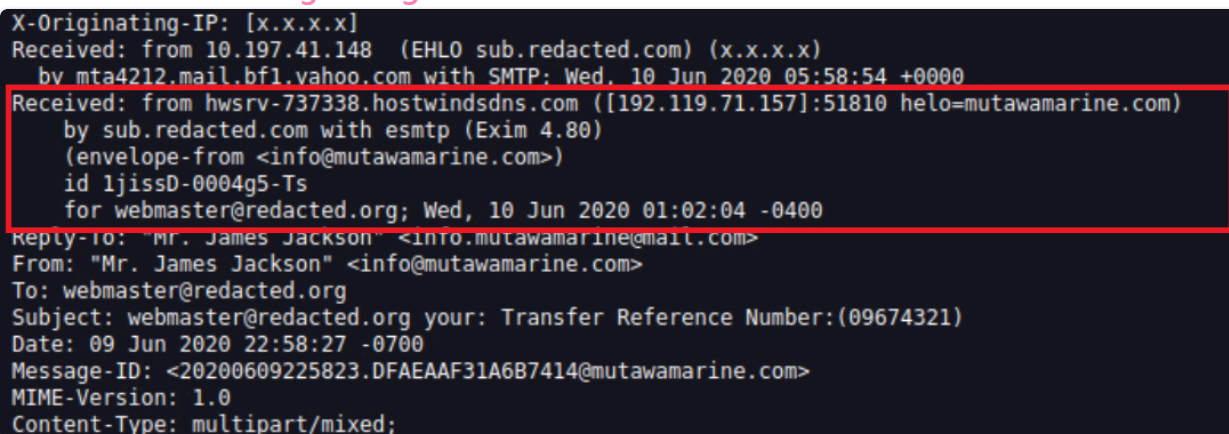
Question Answer & Email analysis

1. What is the **Transfer Reference Number** listed in the email's **Subject** line?
 - Transfer Reference Number: **09674321**
2. What is the display name of the sender?
 - by looking for email header we can find display name of the sender is **Mr. James Jackson**
3. What is the sender's email address?
 - sender's email address : info@mutawamarine[.]com
4. What email address will receive a reply to this email?
 - from email header will find reply to domain is : info.mutawamarine@mail[.]com
5. What is the originating IP address of this email?
 - to know originating IP address should analyzing message source by :
 1. open email by Thunderbird and chose View tab from bar
 2. then click Message Source choice



```
Applications Places System Thu Sep 17, 12:36
Source of file:///home/ubuntu/Desktop/challenge.eml? - Mozilla Thunderbird
File Edit View Help
X-Atlas-Received: from 10.201.192.162 by atlas125.free.mail.bf1.yahoo.com with http; Wed, 10 Jun 2020 05:58:55 +0000
Return-Path: <info@mutawamarine.com>
Received: from x.x.x.x (EHLO sub.redacted.com)
by atlas125.free.mail.bf1.yahoo.com with SMTPs; Wed, 10 Jun 2020 05:58:55 +0000
X-Originating-IP: [x.x.x.x]
Received-SPF: fail (domain of mutawamarine.com does not designate x.x.x.x as permitted sender)
Authentication-Results: atlas125.free.mail.bf1.yahoo.com;
spf=fail smtp.mailfrom=mutawamarine.com;
dmarc=unknown
X-Apparently-To: redacted@yahoo.com; Wed, 10 Jun 2020 05:58:55 +0000
X-MailISG: CA2XOWoLDuMav_xVT1F_0kXM35Y6SWpmP6zsE6LeQRxow4
YjzuEZUwEEJzHhUgBkbpzCq7GFztoIFDbqKMKWunxNYA6aofb6xusqm_FJ
x591PPWdY5Nhw7H_PwB9o9VzmzhbgKs3KzMM9107Uhsjfy5y6rUw_dShjUv
j1RqXZyShqUA_Rced5bTlHpyy8T4LoSfHwM9E4FgJ2W219zg0wK3Ss1zNP
GqFhAYID0v_GOUTdBURWp8as0IPa4KYC7v0e0TvmUEDENPAjmcnfzjpo
geisZs471gDrr3JWxiUMUTyCHHRw9nCczLeggA23t_MdbCZ7qgFqWwv0II
nIXk135mwKec90ZCP1c6tCHAQyFke_030_8VnK_bralt5eq0iGBYmyCV2i
CwhdwDtwYkUIIdgler5ES85mHXsqNvtmP00rjMPdg5X1B7yVSIFaIFsr08
OTEw0w1CkKw4gXhNU4FH4Iub03b9TLvUoX9KLEfx3De15yPTf8xXY7NY_kzA
aCwKtjP4Faet2Mk1Pq5P4BDF.DB6hdMTmCoowuSwouW2M9Yp4eugKNzGrLcf
2KcMR0fFvCKDwXSAHw4tMhKvXSH0KiWwFpXPamnt2c0ckLVpwZMyql8w8W
PlponB1e66yiIqNuVv_vt64152HFh0jNwcuohMo7MA70mMP_0tkdwLUqLS
68AfikwFW3Sppf_ptqTPGNPF2wuAsJiaT7_Q0_4x3khgYrC4jTmxjVBMDVRV
wT0AdlQ716U8Tvp_0AKvevKFmZfz0T0TsGLUQU_w8uZhv_6mKB4sW7Pbhbr
B1RJadC0va2CjicbAC1QnapmSeg5IEExqkhboy8i0Ufz0qkD1a1_tn5nv1Ix0
bCqer07cnAjpN1amFUVc8gJ0345qb6k9L7h7a8TFsv_67Nkr0k_M4_MRZcf
.iuxPcffe2r1oca5FW0g6yof0W0ta51sbW0idg7B_4XR2_6cbg8U139t2v2Y
bgWISohtB1urfrp_b1SANr7fve2Zvzjz4_4PbBt8DevFUB7Pjq0G1Ae_Nx
YpW8pLoFasy1lk4T9f5e5ryqAU.HToIeqVimVa4xwuz1bNvaE7Tsm4m3vepb
```

6. search **X-Originating-IP** header but in this mail not found so will use another way to get originating IP address by check email path through reviewing **Received: from** header
7. should start from down because every email server transmit message add its header so first header will be the **originating IP address**



```
X-Originating-IP: [x.x.x.x]
Received: from 10.197.41.148 (EHLO sub.redacted.com) (x.x.x.x)
by mta4212.mail.bf1.yahoo.com with SMTP; Wed, 10 Jun 2020 05:58:54 +0000
Received: from hwsrv-737338.hostwinddns.com ([192.119.71.157]:51810 helo=mutawamarine.com)
by sub.redacted.com with esmtp (Exim 4.80)
(envelope-from <info@mutawamarine.com>)
id 1jissD-0004g5-Ts
for webmaster@redacted.org; Wed, 10 Jun 2020 01:02:04 -0400
Reply-to: "Mr. James Jackson" <info.mutawamarine@mail.com>
From: "Mr. James Jackson" <info@mutawamarine.com>
To: webmaster@redacted.org
Subject: webmaster@redacted.org your: Transfer Reference Number:(09674321)
Date: 09 Jun 2020 22:58:27 -0700
Message-ID: <20200609225823.DFAEAAF31A6B7414@mutawamarine.com>
MIME-Version: 1.0
Content-Type: multipart/mixed;
```

then the originating IP address is : **192.119.71.157**

8. Investigate the IP address from the previous question Who is the owner of the originating IP?
- the owner of the originating IP can be known by using TI websites like IPinfo or WHOIS and more

192.119.71.157

WHOIS Information

WhoisReverse IPDiagnostics

IP Whois

NetRange:	192.119.64.0 - 192.119.127.255
CIDR:	192.119.64.0/18
NetName:	HOSTWINDS-18-2
NetHandle:	NET-192-119-64-0-1
Parent:	NET192 (NET-192-0-0-0-0)
NetType:	Direct Allocation
OriginAS:	
Organization:	HostPapa (HOSTP-7)
RegDate:	2012-11-12
Updated:	2026-05-13
Comment:	Geofeed https://geofeeds.oniaas.io/geofeeds.csv
Ref:	https://rdap.arin.net/registry/ip/192.119.64.0
OrgName:	HostPapa

7. Run an SPF record check on the Return-Path domain identified in the email headers.
What is the full SPF record for this domain?

1. get domain from Return-Path: header

```
X-Atlas-Received: from 10.201.102.162 by atlas125.free.mail.bf1.yahoo.com with http; Wed, 10 Jun 2020 05:58:55 +0000
Return-Path: <info@mutawamarine.com>
Received: from x.x.x.x (EHLO sub.redacted.com)
  by atlas125.free.mail.bf1.yahoo.com with SMTPs; Wed, 10 Jun 2020 05:58:55 +0000
X-Originating-IP: [x.x.x.x]
Received-SPF: fail (domain of mutawamarine.com does not designate x.x.x.x as permitted sender)
Authentication-Results: atlas125.free.mail.bf1.yahoo.com;
  spf=fail smtp.mailfrom=mutawamarine.com;
```

- to know full SPF record for this domain use [SPF Checker | SPF Surveyor](#) from dmarcian

Your SPF Record

— 1 mutawamarine.com

```
v=spf1 include:spf.protection.outlook.com -all
```

8. Perform a DMARC lookup for the Return-Path domain found in the email headers.
What is the complete DMARC record for this domain?

- use DMARC Checker

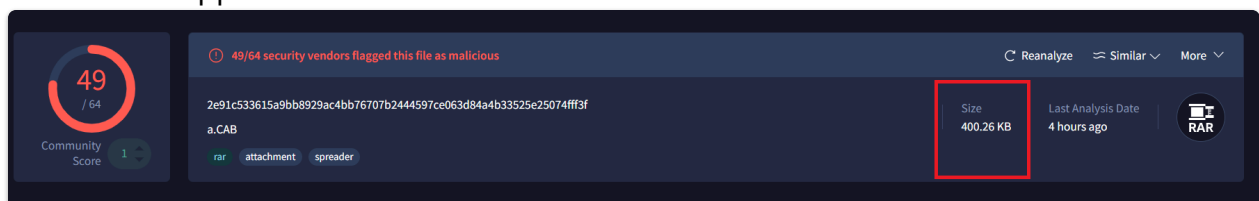
Your DMARC Record

```
v=DMARC1; p=quarantine; fo=1
```

9. What is the file name of the attachment found in the email?
- **SWT_#09674321_PDF.CAB
10. Download the attachment to your virtual environment.
Using the `sha256sum` command, what is the `SHA256` hash of the file?

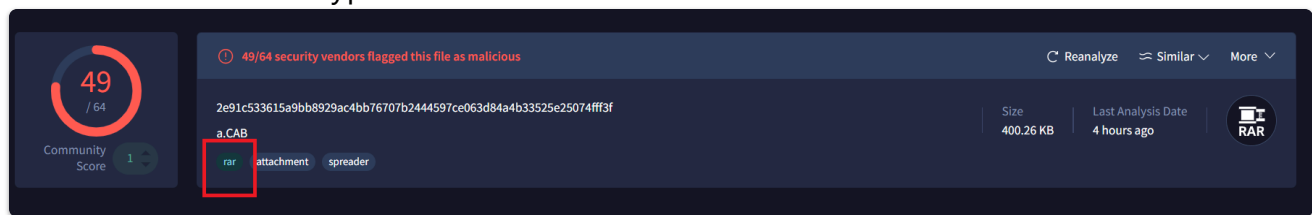
```
ubuntu@tryhackme: ~/Desktop
File Edit View Search Terminal Help
ubuntu@tryhackme:~$ cd Desktop/
ubuntu@tryhackme:~/Desktop$ ls
SWT_#09674321____PDF__.CAB  challenge.eml  mate-terminal.desktop
ubuntu@tryhackme:~/Desktop$ sha256sum SWT_#09674321____PDF__.CAB
2e91c533615a9bb8929ac4bb76707b2444597ce063d84a4b33525e25074fff3f  SWT_#09674321____PDF__.CAB
ubuntu@tryhackme:~/Desktop$
```

11. Investigate the file hash from the previous question using
What is the attachment's file size ?
- take file hash and go to Virus Total and search about file using hash and all information about it will appear



12. Continue your research on the file.

What is the actual file type of the attachment?



IOC table

Indicator	Type	Value
Sender email	Email	info@mutawamarine[.]com
Reply-to	Email	info.mutawamarine@mail[.]com
Originating IP	IP	192.119.71[.]157
Attachment	File	09674321_PDF
SHA256	Hash	2e91c533615a9bb8929ac4bb76707b2444597ce063d84a4b33525e25074f
Actual file type	Type	RAR archive (masqueraded as .CAB)

Summary

- After analyzing this email starting with static header analysis, I found that SPF authentication failed. However, the message was quarantined rather than completely blocked due to the domain's DMARC policy being set to `quarantine`. Analysis of the attachment file using its hash value flagged it on VirusTotal (49/64 vendors), classifying it as a Trojan/Ransomware dropper designed to establish initial access and download a ransomware payload. Ultimately, this email was identified as a malicious Phishing/Business Email Compromise (BEC) attempt impersonating a known customer, originating from an IP address (192.119.71.157) hosted by HostPapa/Hostwinds rather than the legitimate customer's infrastructure.

Recommendations

- 1- Block sender domain/IP on mail gateway.
- 2- Conduct targeted user awareness training on this attack vector.